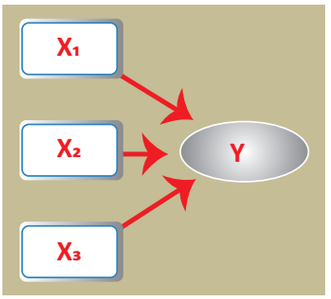
The general form of the multiple linear regression equation is

$Y = a + b_1X_1 + b_2X_2 + b_3X_3 + ... + b_tX_t + e$



Here, Y represents the dependent (response) variable, X_i represents the ith independent variable (regressor), a and b_i are the regression coefficients and e is the error term.

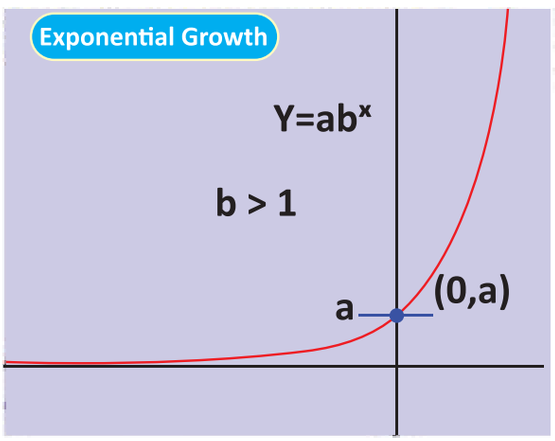
Suppose that price of a product (Y) depends mainly upon three promotional activities such as discount (X_1), instalment scheme (X_2) and free installation (X_3). If the price of the product has linear relationship with each promotional activity, then the relationship among Y and X_1, X_2 and X_3 may be expressed using the above general form as

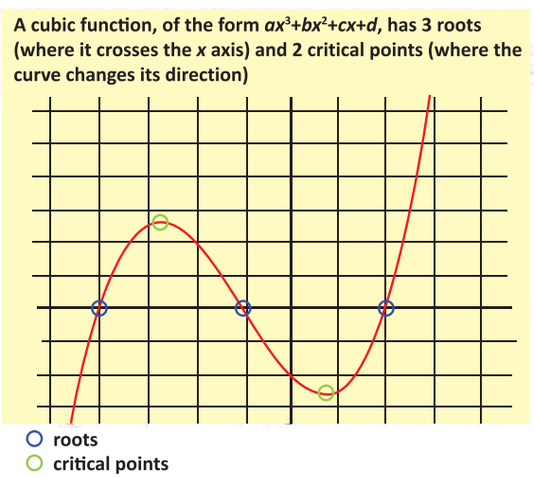
$$Y = a + b_1X_1 + b_2X_2 + b_3X_3 + e\,.$$

These benefits help market researchers / data analysts / data scientists to eliminate and evaluate the best set of variables to be used for building regression models for predictive purposes.

5.1.3 Non-Linear Regression

If the regression is not linear and is in some other form, then the regression is said to be non-linear regression. Some of the non-linear relationships are displayed below.





5.2 USES OF REGRESSION

Benefits of using regression analysis are as follows:

1. It indicates the **significant mathematical** relationship between independent variable (X) and dependent variable(Y). *(i.e)* Model construction
2. It indicates the **strength of impact (b)** of independent variable on a dependent variable.
3. It is used to estimate (interpolate) the value of the response variable for different values of the independent variable from its range in the given data. It means that extrapolation of the dependent variable is not generally permissible.

NOTE

Multiple linear regression and Curvilinear relationships (non-linear regression) are out of the syllabus. Basic information about them are given here, for enhancing the knowledge.

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